Constructing Diversification Constraints

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Aug 1, 2026

1 Overview

When one constructs (or modifies) a long-only portfolio of n securities — a vector $\mathbf{x} \succeq \mathbf{0}$ of notional positions[1] — a diversification mandate will often demand that the largest positions not dominate the book: *the k largest positions may not, together, exceed a fraction p of the portfolio's total notional.*[2] The difficulty in enforcing such a rule inside an optimization is that *which* k positions are the largest depends on $\mathbf{x}$ — the very variable being optimized. One way out is to write down, for every set of k securities, a constraint bounding the sum of those positions; but the number of constraints that one has to write is $\binom{n}{k}$. This becomes combinatorially large very quickly. In this paper we use LP duality to replace these $\binom{n}{k}$ constraints with $O(n)$ linear constraints.

To formalize, let $k \in \mathbf{Z}^+$ with $k \leq n$ throughout, and consider the function, $f : \mathbf{R}^n \to \mathbf{R}$, defined by[3]

$$f(\mathbf{x}) = \sum_{i=1}^{k} x_{[i]}.$$

[1] A bold font is used to denote vectors. We refer to a vector of 0s as $\mathbf{0}$, and likewise we refer to a vector of 1s as $\mathbf{1}$. The mathematical expression, $\mathbf{a} \preceq \mathbf{b}$, is to be interpreted as the statement that every element of the vector $\mathbf{a}$ is less than or equal to the corresponding element in the vector $\mathbf{b}$. Similarly, the expression, $\mathbf{a} \succeq \mathbf{b}$, is the statement that each element of $\mathbf{a}$ is greater than or equal to each corresponding element of $\mathbf{b}$.

[2] Rules of this shape occur in practice; the European UCITS "5/10/40" rule is a well-known relative.

[3] For a given vector, $\mathbf{x} \in \mathbf{R}^n$, create a vector $\mathbf{x}'$ by sorting $\mathbf{x}$ from highest to lowest (ties broken arbitrarily, as the sums below are independent of the tie-breaking rule). Define the notation, $x_{[i]}$ by: $x_{[i]} \equiv x'_i$.

That is, $f(\mathbf{x})$ is the sum of the k largest entries of $\mathbf{x}$. Note that f is *convex*: it is the maximum, over all $\binom{n}{k}$ index subsets S of size k, of the linear functions $\mathbf{x} \mapsto \sum_{i \in S} x_i$. This convexity is the reason the constraint $f(\mathbf{x}) \leq M$ can admit a linear representation at all. We are interested in optimization problems involving a vector $\mathbf{x}$ where the sum of the top k entries is bounded; that is, $f(\mathbf{x}) \leq M$ for a given M. The portfolio mandate is the case $M = p\,\mathbf{1}^T\mathbf{x}$. This bound depends on the variable $\mathbf{x}$ — but only *linearly*, so the constraint (8) below remains linear with M replaced by $p\,\mathbf{1}^T\mathbf{x}$. The function f is also k times the *conditional value-at-risk* (CVaR) of the empirical distribution of the entries of $\mathbf{x}$ at level k/n; the reformulation below is precisely the linearization of Rockafellar and Uryasev [3] in that setting.

This reformulation is classical — it appears in Ogryczak and Tamir [1]; see also Boyd and Vandenberghe [2] — but the derivation below is self-contained. The route is a *change of representation*, made three times: the sorted sum becomes a 0–1 integer program; the integer program becomes a continuous linear program; and the linear program is traded for its *dual*. Each trade is forced by a defect of *shape* in the current description; the table at the end of the paper collects the ladder.

2 The Function f as an Optimization Problem

We wish to characterize $f(\mathbf{x})$ as the value of an optimization. We claim that for a given $\mathbf{x}$, $f(\mathbf{x})$ equals the optimal value of the following constrained *discrete* optimization problem:

$$\max_{\mathbf{y} \in \{0,1\}^n} \quad \mathbf{y}^T\mathbf{x} \tag{1}$$

$$\text{s.t.} \quad \mathbf{y}^T\mathbf{1} = k \tag{2}$$

This is $f(\mathbf{x})$ written as a 0-1 integer program (IP). Although succinct, this optimization still requires examining $\binom{n}{k}$ candidate solutions.

Claim. The discrete optimization problem is equivalent to the following continuous problem:[4]

$$\begin{aligned}
\max_{\mathbf{y}\in\mathbf{R}^n} \quad & \mathbf{y}^T\mathbf{x} \\
\text{s.t.} \quad & \mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1} && (3) \\
& \mathbf{y}^T\mathbf{1} = k && (4)
\end{aligned}$$

Proof. Since its feasible set contains the discrete problem's feasible set (every $\mathbf{y} \in \{0,1\}^n$ with $\mathbf{y}^T\mathbf{1} = k$ also satisfies $\mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1}$), the optimal value of the continuous problem is at least as large as the optimal value of the original discrete problem. We now show the reverse. For any optimal solution $\mathbf{y}^*$ of the continuous problem, we construct a $\mathbf{y}' \in \{0,1\}^n$ satisfying (2) and yielding the same value of the objective (1) as $\mathbf{y}^*$.

If all components of $\mathbf{y}^*$ are integers, by (3) they are each 0 or 1; in this case take $\mathbf{y}' = \mathbf{y}^*$.

Otherwise, consider the components of $\mathbf{y}^*$ that are non-integer; that is, have values strictly between 0 and 1. By (4), $\sum_i y_i^* = k$ is an integer; the integer-valued components contribute an integer to this sum, so the non-integer components must sum to an integer too. Consequently, there must be more than one such component. It must also be the case that amongst these non-integer components the corresponding entries of $\mathbf{x}$ are the same. Otherwise, there would be indices i, j with $y_i^*, y_j^* \in (0,1)$ and $x_i > x_j$; shifting a small amount $\epsilon > 0$ from y_j^* to y_i^* preserves the constraints (3) and (4) (both components are strictly between 0 and 1, so a small shift stays within the bounds) and increases the objective by $\epsilon(x_i - x_j) > 0$ — contradicting the optimality of $\mathbf{y}^*$. Therefore, all of the entries of $\mathbf{x}$ associated with non-integer components of $\mathbf{y}^*$ have the same value. As previously stated, the sum of these non-integer values is an integer, which is positive and necessarily less than the number of non-integer components in $\mathbf{y}^*$ (each such component lies strictly between 0 and 1). Let j denote this integer sum. Pick any j of these non-integer values, and let J be the set of their index values in $\mathbf{y}^*$. Let $\tilde{J}$ be the set of indices corresponding to the remaining non-integer values of $\mathbf{y}^*$.

Now, create a new vector $\mathbf{y}' \in \{0,1\}^n$ and set its values in the following way: Set the values of $\mathbf{y}'$ corresponding to the indices in J to be 1. Set the values of $\mathbf{y}'$ corresponding to the indices in $\tilde{J}$ to 0. Set the rest of the components of $\mathbf{y}'$ to the corresponding values from $\mathbf{y}^*$. By our construction, $\mathbf{y}'^T\mathbf{1} = k$. The components of $\mathbf{y}'$ inherited from $\mathbf{y}^*$ were already integers, and by (3) every integer component of $\mathbf{y}^*$ is 0 or 1. It follows that $\mathbf{y}' \in \{0,1\}^n$ and satisfies the constraint of the discrete problem, (2). The objective value is also preserved: let x_c denote the common value of the entries of $\mathbf{x}$ on $J \cup \tilde{J}$. The contribution to the objective

[4] The objective $\mathbf{y}^T\mathbf{x}$ is continuous and the feasible set is closed and bounded (a compact polytope in $\mathbf{R}^n$); the supremum is therefore attained, so we may write "max" instead of "sup".

from those indices equals $x_c \sum_{i \in J \cup \tilde{J}} y_i^* = x_c\, j$, which is exactly the contribution from $\mathbf{y}'$ on the same indices. Therefore, $\mathbf{y}'$ is in the feasible set of the discrete problem and attains the optimal value of the continuous problem; that is, the optimal value of the discrete problem is at least the optimal value of the continuous problem. Consequently, the two optimization problems are equivalent.

Remark. The equivalence is no accident: the vertices of the polytope $\{\mathbf{y} : \mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1},\ \mathbf{y}^T\mathbf{1} = k\}$ are exactly the 0–1 vectors with k ones, and a linear objective always attains its maximum at a vertex. The proof above has the advantage of being self-contained.

The continuous problem above is equivalent to the following minimization:

$$\min_{\mathbf{y} \in \mathbf{R}^n} \quad -\mathbf{y}^T\mathbf{x} \tag{5}$$

$$\text{s.t.} \quad \mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1} \tag{6}$$

$$\mathbf{y}^T\mathbf{1} = k \tag{7}$$

Treating $\mathbf{x}$ as a constant, this is a *convex* optimization problem.

3 A Dual Description of the Optimization

Pause to ask what shape of description we *want*. Every description of $f(\mathbf{x})$ so far is a *maximum*. A maximum is the wrong shape to sit inside the constraint $f(\mathbf{x}) \leq M$: to certify that a maximum is at most M one must check *every* candidate — for us, all $\binom{n}{k}$ of them (the last section makes this precise). What we want is a description of $f(\mathbf{x})$ as a *minimum*: to certify that a minimum is at most M it suffices to exhibit *one* feasible point with objective value at most M. Duality is precisely the machine that converts the one into the other.

Since the problem described by (5)–(7) is *convex*, the value of its solution is the same as the value of its associated *dual* problem.[5]

[5] Normally one needs to show that a convex problem satisfies Slater's condition in order to invoke strong duality. For linear programs, strong duality holds whenever the primal is feasible and bounded – no Slater check needed. The bilinear term $-\mathbf{y}^T\mathbf{x}$ becomes linear in $\mathbf{y}$ once we fix $\mathbf{x}$ as a parameter.

To form the dual problem we need the Lagrangian,[6] which is:[7]

$$L(\mathbf{y}, \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = -\mathbf{y}^T\mathbf{x} - \boldsymbol{\lambda}_1^T\mathbf{y} + \boldsymbol{\lambda}_2^T(\mathbf{y} - \mathbf{1}) + \nu(\mathbf{y}^T\mathbf{1} - k).$$

Here $\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2 \in \mathbf{R}^n$ with $\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2 \succeq \mathbf{0}$ (sign constraints from the two inequality systems) and $\nu \in \mathbf{R}$ free (from the equality) are the *dual* variables.

Define g by:

$$g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = \inf_{\mathbf{y}} \; L(\mathbf{y}, \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}).$$

Substituting for L this becomes

$$g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = \inf_{\mathbf{y}} \Big[\mathbf{y}^T\big(-\mathbf{x} - \boldsymbol{\lambda}_1 + \boldsymbol{\lambda}_2 + \nu\mathbf{1}\big) - \mathbf{1}^T\boldsymbol{\lambda}_2 - \nu\, k \Big].$$

The dual problem is then

$$\max_{\substack{\boldsymbol{\lambda}_1 \succeq \mathbf{0} \\ \boldsymbol{\lambda}_2 \succeq \mathbf{0} \\ \nu}} \quad g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}).$$

Which is[8]

$$\begin{aligned} \max_{\substack{\boldsymbol{\lambda}_1 \succeq \mathbf{0} \\ \boldsymbol{\lambda}_2 \succeq \mathbf{0} \\ \nu}} \quad & -\mathbf{1}^T\boldsymbol{\lambda}_2 - \nu\, k \\ \text{s.t.} \quad & -\boldsymbol{\lambda}_1 + \boldsymbol{\lambda}_2 + \nu\mathbf{1} = \mathbf{x} \end{aligned}$$

This is equivalent to[9]

$$\begin{aligned} \max_{\substack{\boldsymbol{\lambda} \succeq \mathbf{0} \\ \nu}} \quad & -\nu\, k - \mathbf{1}^T\boldsymbol{\lambda} \\ \text{s.t.} \quad & \mathbf{x} \preceq \boldsymbol{\lambda} + \nu\mathbf{1} \end{aligned}$$

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[6] In companion papers — "Properties of a Linear Unbiased Minimum Variance Estimator", and the minimum-norm lemma of "Singular Transformations of Probability Density Functions" — Lagrange multipliers are scaffolding: introduced to locate a minimizer, then discarded. Here the multipliers are the *deliverable*: $\boldsymbol{\lambda}$ and ν survive into the final answer as the new decision variables of the next section, and ν will even acquire a meaning — see the Remark below.

[7] $\mathbf{x}$ is a parameter (constant for the optimization) – separated from the optimization variables by a semicolon.

[8] Note that g is $-\infty$ unless the coefficient of $\mathbf{y}$ is the zero vector. Consequently, the maximum must necessarily occur where that coefficient is zero.

[9] We can remove n variables and the n equality constraints by eliminating $\boldsymbol{\lambda}_1$. The equality $-\boldsymbol{\lambda}_1+\boldsymbol{\lambda}_2+\nu\mathbf{1} = \mathbf{x}$ together with $\boldsymbol{\lambda}_1 \succeq \mathbf{0}$ is equivalent to $\boldsymbol{\lambda}_2 + \nu\mathbf{1} \succeq \mathbf{x}$. Since there is now only one $\boldsymbol{\lambda}$, we relabel $\boldsymbol{\lambda}_2$ as $\boldsymbol{\lambda}$.

Negating the objective converts the max to a min:

$$\begin{aligned} \min_{\boldsymbol{\lambda},\nu} \quad & \nu\, k + \mathbf{1}^T \boldsymbol{\lambda} \\ \text{s.t.} \quad & \mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1} \\ & \boldsymbol{\lambda} \succeq \mathbf{0} \end{aligned}$$

Remark. The dual has a useful closed form. For fixed ν, the best choice of $\boldsymbol{\lambda}$ is clearly $\lambda_i = (x_i - \nu)_+ \equiv \max(x_i - \nu,\, 0)$, so the dual value is

$$f(\mathbf{x}) = \min_{\nu \in \mathbf{R}} \Big[\, k\nu + \sum_{i=1}^{n} (x_i - \nu)_+ \Big],$$

and the minimum is attained at $\nu^* = x_{[k]}$, the k^{th} largest entry of $\mathbf{x}$. Indeed, at $\nu = x_{[k]}$ the bracket evaluates to $k\, x_{[k]} + \sum_{i=1}^{k} \big(x_{[i]} - x_{[k]}\big) = \sum_{i=1}^{k} x_{[i]} = f(\mathbf{x})$, which exhibits the optimal dual point directly, without appeal to LP theory. The multiplier ν acts as a threshold, and $\boldsymbol{\lambda}$ pays for the excess of each entry above it.

The Remark buys more than an interpretation. Weak duality is here a one-line computation: if $\mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1}$ with $\boldsymbol{\lambda} \succeq \mathbf{0}$, and S is an index set of the k largest entries of $\mathbf{x}$, then

$$f(\mathbf{x}) = \sum_{i \in S} x_i \;\le\; \sum_{i \in S} (\lambda_i + \nu) \;=\; \nu\, k + \sum_{i \in S} \lambda_i \;\le\; \nu\, k + \mathbf{1}^T \boldsymbol{\lambda}.$$

So every feasible $(\boldsymbol{\lambda}, \nu)$ bounds $f(\mathbf{x})$ from above, and the point exhibited above attains the bound. Together the two lines prove — elementarily — the strong duality that the next section uses.

A calibration. Take $n = 4$, $\mathbf{x} = (3, 1, 2, 2)^T$, and $k = 2$. Sorting gives $f(\mathbf{x}) = 3+2 = 5$. The recipe above says $\nu^* = x_{[2]} = 2$ and $\boldsymbol{\lambda}^* = \big((3-2)_+,\ (1-2)_+,\ (2-2)_+,\ (2-2)_+\big)^T = (1, 0, 0, 0)^T$, with dual value $\nu^* k + \mathbf{1}^T \boldsymbol{\lambda}^* = 4 + 1 = 5$. The new machine reproduces the answer the reader already owns by sorting — which is how one learns to trust a new tool. The example also realizes the tie that the equivalence proof of Section 2 labors over: since $x_3 = x_4 = 2$, the continuous problem has optimal solutions $\mathbf{y}^* = (1, 0, t, 1-t)^T$ for every $t \in [0, 1]$, all of value 5; the proof's rounding sends each of them to $(1, 0, 1, 0)^T$ or $(1, 0, 0, 1)^T$.

4 Characterization of f as $O(n)$ Linear Constraints

We now apply the results of the previous sections to the following problem: Add constraints to an optimization problem that includes a *variable* $\mathbf{x} \in \mathbf{R}^n$ so that $f(\mathbf{x}) \le M$ for a given

M.

We claim that one can bound $f(\mathbf{x})$ by adding $n+1$ new variables (a vector $\boldsymbol{\lambda} \in \mathbf{R}^n$ and a scalar $\nu \in \mathbf{R}$), along with the following:

$$\nu\, k + \mathbf{1}^T \boldsymbol{\lambda} \leq M \tag{8}$$

$$\mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1} \tag{9}$$

$$\boldsymbol{\lambda} \succeq \mathbf{0} \tag{10}$$

Why? Because for any $(\boldsymbol{\lambda}, \nu)$ satisfying (9) and (10), $\nu\, k + \mathbf{1}^T \boldsymbol{\lambda}$ is an upper bound on $f(\mathbf{x})$ – this is weak duality. Consequently, we have the inequality: $f(\mathbf{x}) \leq \nu\, k + \mathbf{1}^T \boldsymbol{\lambda} \leq M$. The only concern remaining is that there may be a gap between the values of $\nu\, k + \mathbf{1}^T \boldsymbol{\lambda}$ and $f(\mathbf{x})$ – making (8) too restrictive. But this is not the case: by strong duality (proved elementarily in the Remark of the previous section), the minimum of $\nu\, k + \mathbf{1}^T \boldsymbol{\lambda}$ over $\boldsymbol{\lambda}, \nu$ (subject to (9) and (10)) equals $f(\mathbf{x})$. So there always exist $\boldsymbol{\lambda}, \nu$ feasible for (9) and (10) with $\nu\, k + \mathbf{1}^T \boldsymbol{\lambda} = f(\mathbf{x})$, and (8) is satisfiable for $\mathbf{x}$ exactly when $f(\mathbf{x}) \leq M$ – the new constraints introduce no restrictions on $\mathbf{x}$ beyond $f(\mathbf{x}) \leq M$ itself.

One might ask why we couldn't do exactly the same procedure with the "simpler" characterization we had with equations (5)–(7), rather than going through so much effort with a dual problem. It would require the same number of variables and constraints.

We could try that; however, note that once we take the characterization of a bound on a "given" $\mathbf{x}$ and place it into an optimization problem, that "given" $\mathbf{x}$ is no longer a constant. It is now part of the optimization. The primal characterizes $f(\mathbf{x})$ as a *maximum* over $\mathbf{y}$; to enforce $f(\mathbf{x}) \leq M$ we would need $\mathbf{y}^T\mathbf{x} \leq M$ for *every* feasible $\mathbf{y}$ – which is exactly the original combinatorial blow-up. The only way to collapse it into a finite set of constraints is to encode the LP's optimality conditions, which introduce bilinear terms in the (now) *variable* $\mathbf{x}$ and the new *variable* $\mathbf{y}$. These terms are decidedly *non-linear*; in fact, they are not even *convex*.

In contrast to this "simpler" characterization and the original naïve combinatorial approach, we have shown that by examining an associated dual problem, $f(\mathbf{x})$ can be bounded by adding $(n+1)$ new variables and $(2n+1)$ *linear* constraints.

Return, finally, to the portfolio mandate that opened this paper. To require that the top k positions of a long-only portfolio not exceed the fraction p of its total notional, adjoin the variables $\boldsymbol{\lambda} \in \mathbf{R}^n$ and $\nu \in \mathbf{R}$ to the portfolio problem and impose (8)–(10) with $M = p\,\mathbf{1}^T\mathbf{x}$:

$$\nu\, k + \mathbf{1}^T \boldsymbol{\lambda} \leq p\,\mathbf{1}^T\mathbf{x}, \qquad \mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1}, \qquad \boldsymbol{\lambda} \succeq \mathbf{0}.$$

Every constraint is linear in $(\mathbf{x}, \boldsymbol{\lambda}, \nu)$, so any LP or QP solver accepts them unchanged, and the mandate is enforced exactly — no sorting, and no enumeration of $\binom{n}{k}$ subsets.

In summary, the paper has been one number wearing five descriptions, each traded for the next because of a defect of shape:

description of $f(\mathbf{x})$	shape	defect forcing the next rung
sum of the k largest entries	a sort	a sort is not a constraint
$\max_{\lvert S\rvert=k} \sum_{i \in S} x_i$	max of $\binom{n}{k}$ linear pieces	too many pieces
0–1 program (1)–(2)	max over a discrete set	combinatorial search
linear program (3)–(4)	max over a polytope	a max needs *every* $\mathbf{y}$ checked
dual program (8)–(10)	min	none — *one* witness certifies

The same strategic move — change the representation until its shape fits the intended use — carried each rung of the ladder.

References

[1] W. Ogryczak and A. Tamir, Minimizing the sum of the k largest functions in linear time, *Information Processing Letters* **85** (2003), 117–122.

[2] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.

[3] R. T. Rockafellar and S. Uryasev, Optimization of conditional value-at-risk, *Journal of Risk* **2** (2000), 21–41.